07/26 Experiment #4

Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer..

Aim:

To design a simple topology and configure with one router, two switches and PCs using cisco packet tracer.

Steps:

1. Build the topology

(i) Add End Devices: Navigate to End devices in the bottom-left panel. Drag four PCs onto the logical workspace. Place PC0 and PC1 on the left side (LAN1). place PC2 and PC3 on the right side (LAN2)

(ii) Add switches: Go to Network devices. Switches choose the Cisco 2960 switch. Drag two switches onto the workspace (switch0 for LAN1; switch1 for LAN2)

(iii) Add a Router: Go to Network devices > Routers. Drag a Cisco 4331 or 2911 Router and place it in the centre, above both sides.

(iv) cable the Devices: Select the connections and icons and choose the copper straight through cable.

- Connect PC0 (Fast Ethernet) to switch0
- Connect PC1 (Fast Ethernet 0) to switch0
- Connect PC2 (Fast Ethernet 0) to switch1
- Connect PC3 (Fast Ethernet 0) to switch1
- connect switch0 to Router0
- connect switch1 to Router0

2. configure the PCs.
(i) click the PC → Desktop → IP configuration
(ii) Select static
(iii) Enter the following details

PC	IP Address	Subnet mask	Default gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.20.10	255.255.255.0	192.168.10.1
PC3	192.168.20.11	255.255.255.0	192.168.10.1

3. Configure the Router interfaces;
(i) Click Router 0 → CLI
(ii) If prompted, type no and press Enter
(iii) Enter the following commands.
(Use the interface names for your router)

enable
configure terminal
interface gigabitEthernet 0/0
IP address 192.168.10.1 255.255.255.0
no shut down
exit
interface gigabitEthernet 0/1
IP address 192.168.20.1 255.255.255.0
no shutdown
exit

Write manually
Note: After to shutdown, the interface lights change from red to green, indicating the interface are active.

4. Verify connectivity:
(i) on pc, goto desktop → command prompt
(ii) ping the default gateways.
(iii) ping a PC on the other network
ping 192.168.20.10

Result:
The network working correctly. The first ping timeout due to ARP resolution pings should receive reply from 192.168.20.10 confirming successfull communication between LAN1 to LAN2 through the routers.